Interview Questions

The questions put to the interviewees

Part A: Women and other minority communities in STEM at IISER Kolkata

Interview 1: Dr. Rahul Banerjee, Dean of Faculty Affairs (DoFA)

How faculty are recruited

Q1. How are faculty recruited at IISER Kolkata? How does the entire process work?

Q2. How long does the process usually take?

Policy on women in hiring

Q3. Does the government set any target or reservation for women in faculty hirings? Are there any other types of government reservations for the number of female students or faculty?

Few women apply

Q4. How many women apply, and in which departments or specialisations are applications from women fewest?

Q5. Why do you think women apply less?

Q6. Is anything being done to encourage women to aspire to faculty posts?

IISER Kolkata, TVM and Pune

Q7. Why is IISER Kolkata's share of women faculty, about 14% since 2019, lower than at TVM and Pune (22 to 24%)?

Q8. What, in your opinion, is the main reason for the low number of women faculty across India, especially in the INIs?

Q9. Why are the numbers of faculty from SC, ST and OBC communities so low, even with a reservation policy? Why does the pattern show only slight changes after 2019 across India?

Biology and Chemistry versus Physics and Maths

Q10. Why are women fewer in Physics and Maths than in Biology and Chemistry?

Q11. Who should we talk to about this?

Pace of hiring and promotions

Q12. We notice an anomaly: although, across India, the number of women faculty in senior positions (like professors) tends to be low, at IISER Kolkata we notice relatively more women in senior posts. What could be the reasons for this?

Q13. Is there data on how many women (and how many SC, ST and OBC candidates) applied, were offered a position and joined? Is there domicile data for faculty?

Interview 2: Dr. Rangeet Bhattacharyya, Dean of Academic Affairs (DoAA)

Verifying our data

Q1. Is our data on admissions and gender correct?

Q2. Why do you think the share of women is so low?

Q3. Why does the share of women vary so much across departments, and is it performance or something else?

Admission and enrolment data

Q4. What do the official admission records show by gender, and is this pattern specific to one year?

Q5. What are the male to female ratios by department across the institute?

Q6. How many women are there among the faculty in Physics and Maths?

Trend over the years

Q7. What have the male to female ratios in Maths been across the BS-MS batches 14 MS to 21 MS?

Q8. How steady is the BS-MS intake ratio, and how does Chemistry compare with it?

Q9. What is the picture for the PhD programme?

Reasons

Q10. What can be the actual and perceived reasons for the low share of women?

Q11. Is there CGPA data by gender so that we can test the hypothesis? Are there any alternative ways to check if the data holds across different subjects for the first two years before major selection?

Q12. If Earth Sciences is not a mathematical subject, why does it also have a poor ratio?

Analysis he recommended

Q13. What analysis would you recommend we run, for undergraduates and for PhD students?

Q14. We found PhD is slightly better than BS-MS (about 32% against 27%). Which departments do you expect to be better?

Use of the data

Q15. May we use the anonymised gendered CGPA data and the subject-wise SGPA data?

Why women drop off later

Q16. Why do women drop off later, from your personal experience, during the PhD and among faculty?

SC, ST and OBC faculty

Q17. Why are SC, ST and OBC faculty numbers so low?

Q18. What was the roster rule before and after 2019?

Q19. How has the number been changing since 2019?

Interview 3: Dr. Anindita Bhadra (DBS)

Opening

Q1. Why is the share of women low among both students and faculty at IISER Kolkata?

Q2. Why are women better represented in the biological sciences than in physics and maths (DBS has 9 women to 17 men)?

Q3. What is your view on the early findings of our survey (scholarships and cost, families' worries, the lack of role models)?

Q4. Who were your role models, and what challenges have you faced as a woman in STEM?

Fewer female students

Q5. Does the share of students from a home state explain the share of girls?

Q6. What other factors could be responsible?

Fewer female faculty

Q7. Why are there fewer female faculty? (Follow-up: how do location and the spouse's job come into it?)

Biology versus Physics and Maths

Q8. Why are women better represented in Biology than in Physics and Maths?

Q9. Do the US findings about girls fearing being called nerds apply in India?

Q10. Do families' attitudes to a daughter being the only girl in a class play a part?

Role models

Q11. Do role models matter?

Q12. Does it matter to students whether the professor is a woman?

Her experience

Q13. What has been your experience as a woman in STEM, as a student and in your job?

Q14. May we reference your article on this?

Resources

Q15. Are there resources you would point us to?

Interview 4: Dr. Malancha Ta (DBS)

Opening

Q1. Why is the share of women faculty at IISER Kolkata low, and why is the share of women students low?

Q2. What do you make of our survey result that around 1 in every 2 responses said role models would have made a difference?

Q3. Why does the ratio fall from Biology to Chemistry to Earth Sciences to Physics to Maths?

Q4. Do role models matter, and what have you faced as a woman in STEM?

Few female faculty

Q5. What explains the low share of women faculty?

Q6. Is the two-body problem the main reason? (Follow-up: how does the area around the campus affect it?)

Few female students

Q7. Why are there few female students? Does home state matter?

Q8. What are the other factors that might matter?

Role models and leadership

Q9. Does a lack of role models explain the low numbers?

Q10. What do the numbers on women directors and deans (about 11%) tell us?

Where the pipeline changes

Q11. Girls do well in school boards, so where does the pipeline change?

Q12. What advice do you have for women students?

Biology versus Physics and Maths

Q13. Why does the ratio fall from Biology to Physics and Maths?

Q14. Does the claim that there is a brain-based reason hold any water?

Q15. Do even educated parents steer girls toward biology unknowingly?

Women in decision-making

Q16. Do women make decisions differently?

Q17. Follow-up: Does the improvement come from girls being inspired, or from the decisions women make?

Her own experience

Q18. Have you felt a disadvantage as a woman in education and now in a position of authority?

Q19. What is your advice to younger women?

Her views on our suggestions

Q20. What do you think of our idea of tuition waivers for top-performing women?

Q21. Would safe transport at night for women be useful?

Q22. What do you think of the hostel for female PhD students, which the DoFA said is being sped up?

Q23. What would help faculty with children?

The data discussion

Q24. What do you make of our undergraduate shares (Kolkata 27.2%, Pune 27.6%, Berhampur 43.7%) given that all IISERs use the same entry channel?

Q25. Is there data on how many boys and girls apply through IAT?

Q26. What other comparisons would you suggest (for example IACS, state-level data), and is NIRF data enough without an RTI?

Interview 5: Dr. Anuradha Bhat (women in STEM)

Opening

Q1. What do you make of the patterns in our reports on women at the IISERs (Tirupati and TVM high, Kolkata and Pune low, Mohali increasing, more women in biology)?

Q2. Why is the proportion of women higher in Biology and Chemistry than in Physics and Maths?

Q3. Why are there fewer women at IISER Kolkata, and do role models matter?

Ranking, preference and the share of women

Q4. After her suggestion that ranking might have a bearing; follow-up: how can we explain that women's share runs almost opposite to institute ranking and student preference?

Q5. Where does a gap at admission come from when the test is national with no gender reservation?

Why fewer women reach the top ranks

Q6. Why might fewer women reach the top ranks?

Q7. Follow-up: Could school type (government versus private) be relevant?

Across departments

Q8. Why does the proportion of women decrease from Biology to Chemistry to Physics to Maths?

Q9. Is there research on how girls are conditioned from a young age?

Regional and career-path factors

Q10. After her suggestion of regional differences: Do attitudes in West Bengal and the South differ, and how does Mohali compare?

Q11. Follow-up: Should we look at the number of Bengalis in the region?

Female faculty and PhD students

Q12. Why are there few female faculty at IISER Kolkata (location, the two-body problem)?

Q13. What campus facilities would help female faculty?

Q14. Her question to us: Are the low percentages reliable, given the small absolute number of faculty? Try to compare absolute numbers with other institutes.

Role models

Q15. Does a female role model make a difference?

Q16. Who is your own role model?

Q17. Has the trend changed over the last 20 years?

Interview 6: Dr. Neelanjana Sengupta, ADoFA (DBS)

Opening

Q1. What do you make of the patterns in our reports (Tirupati and TVM high, Kolkata and Pune low, more women in biology)?

Q2. Why are there fewer women at IISER Kolkata, and do role models matter?

Few female faculty

Q3. Why are there so few female faculty?

Q4. Do recruitment standards and the number of applications from women play a part?

Q5. Is this the traditional leakage of talent along the pipeline, and what causes it?

Q6. What part does the family and the two-body problem play, and is there an immediate solution?

What the institute is doing

Q7. Does the institute have any policy to prefer women when two applicants are equally good?

Q8. What is the institute doing to increase numbers?

Biology versus Physics and Maths

Q9. Why does the ratio fall from Biology to Physics and Maths?

Q10. Is it about talent for maths and physics?

Q11. What does the pipeline need?

Student numbers

Q12. How do you read our batch figure of about 75 women out of 299 students?

Role models

Q13. Do role models matter, and who were yours?

Challenges and tokenism

Q14. How do you feel about conferences and committees that include women for representation?

Q15. May we quote you on this without naming anyone?

Interview 7: Dr. Koel Das (HOD, Mathematics and Statistics)

Opening

Q1. Why are female faculty numbers low?

Q2. Do students approach you about this?

Q3. Do female role models make a difference?

Q4. What have you faced that might also stop people from applying to IISER Kolkata as faculty or students?

Q5. Why is the ratio of female faculty more concentrated in the senior ranks than in the lower ranks?

Recruitment of women faculty

Q6. Why has recruitment of women faculty been low, and why are most women faculty senior?

Q7. How has recruitment been since the early years compared with other IISERs such as Pune?

Location and the two-body problem

Q8. Does the location matter for faculty families?

Q9. Is the two-body problem applicable to IISER Kolkata?

Hiring policy

Q10. Is there a reservation for women, or a preference when candidates are equal?

Physics and Maths

Q11. Why are women fewest in Physics and Maths?

Q12. Do the Western studies on girls being put off maths apply in India?

Q13. Why do women faculty and PhD students drop out?

Q14. How has the number of female PhD students in maths changed recently?

Q15. Does the same pattern show among students?

Decision-making roles

Q16. Have you faced problems inside the institute or outside it (for example when presenting for grants)?

Q17. How can more women be brought into decision-making?

Q18. What do you make of the national figures on women directors and heads of institutions we showed you, and of women deans at IISER Kolkata?

Q19. Should this be enforced?

Undergraduate versus PhD share

Q20. Why is the undergraduate share lower than the PhD share?

Her suggestions

Q21. What would you suggest the institute do (incentives for women, hostels for female PhD students, hiring spouses, more women in decision-making)?

Role models

Q22. Did role models shape your career?

Q23. What did you think of the survey result that many female students said it matters if their professors are women?

Q24. Do female PhD students talk to you more easily?

Q25. What are your views on having more women in positions of power?

Interview 8: Dr. Rumi De (DPS)

Opening

Q1. Why is IISER Kolkata's share of women students (about 27.2%, between 26 and 28% for a long time) below the national figure (49 to 52%) and the INI average?

Q2. Why are the student and faculty shares so low at IISER Kolkata?

Q3. What do you make of our home-state data, our rank and preference correlation and our survey results?

Q4. What do you recommend for our survey of women in the nearby villages?

Why fewer women choose IISER Kolkata

Q5. Is it about IISER Kolkata itself, given that Pune is also among the most preferred and has few women?

Q6. How do parents' concerns affect where daughters go?

Q7. Why do Tirupati and TVM do better?

Home state and nearby states

Q8. Does home state explain the numbers?

Q9. Should nearby states be included, and can this be tested with our data?

The rank and preference correlation

Q10. How do you explain that the most preferred IISERs have the worst gender ratios?

Q11. What is the cause, and how does it compare in ordinary colleges?

Q12. We hesitated to present this because it seemed to show girls falling behind. What is your view?

Q13. Is this a symptom of something deeper?

Performance once at IISER

Q14. We observe that women perform either slightly better or as well as boys once they are here. Can we make any claims from this data?

Q15. What does this say about "capability"?

Q16. Do gender differences on campus come from peers or professors?

Female faculty

Q17. Why is the faculty share about 14% when the entry level is 27%?

Q18. How does the long academic path interact with family responsibility?

Q19. Is there a policy that credits a career break? Should it exist?

Q20. What government support exists for women returning after a break?

Q21. What role does motivation play?

Equity and role models

Q22. Should there be a jump start so people reach an equal platform?

Q23. How can stereotypes about girls and maths be challenged?

Q24. Do women role models help?

Her own experience

Q25. What do students need, and what do they approach you about?

Q26. Where can students go for support?

Interview 9: Dr. Rumi De, follow-up to Interview 8: local NGOs and skills training

Questions

Q1. Our survey of the women in the neighbouring villages showed interest in further education, especially hands-on skills. Is there a way to address this?

Q2. Whom should we approach, and what kind of training would fit (practical skills such as sewing and computer use, and adult literacy classes)?

Part B: Fisherfolk and fish ecology

Interview 10: Mohadep Dutta, fisherman (Part 1)

Questions (translated and asked in Bengali)

Q1. Can fishermen tell male and female fish apart (prawn, elish)?

Q2. Has the ratio of males to females changed over time?

Q3. How has the quantity of fish changed since you started fishing at the age of 7? Do you know why the catch has decreased?

Q4. What is the state of the river? How does it impact your livelihood?

Q5. What other difficulties do you face in fishing?

Q6. Does anyone help you? Are there any regulatory bodies you can approach for help?

Q7. Who would you tell about this, and what happens when you raise it?

Q8. How do the rules about small fish affect your work?

Interview 11: Gobinda Shikdar and Sukuma Sarkar, fishermen (Part 2)

Questions (translated and asked in Bengali)

Q1. How long have you been fishing, and what has fish meant to your family?

Q2. Why have the fish decreased, and what do you think causes the pollution?

Q3. Could you use Ganga water before, and can you now?

Q4. Are there any places apart from the Ganga where you have been fishing? What about the ocean?

Q5. Does fishing still give you a livelihood? How do you earn, and how much from a night of fishing?

Q6. What do you mostly catch, and can you tell male and female hilsa apart?

Q7. What do you think of the government's rules on catching small fish?

Q8. How have development, land, food, fish, the cost of living, health and education changed over your lifetime?

Interview 12: Dr. Anuradha Bhat (fish ecology)

Questions

Q1. How can the gap between ecology-based policy (catch only large fish) and livelihood-based reality be addressed?

Q2. How does the government handle pollution regulation, and who bears the enforcement?

Q3. Does action need to be taken on both sides, industry and fishermen? What could be done to address the gap in policy and to help the fishermen?

Q4. Could inland freshwater aquaculture (pisciculture) replace reliance on river water, and which species should be reared?

Q5. Can you describe how aquaculture farms work?

Q6. Is there a visible, practical marker to tell male and female fish apart, since vitellogenin is only a chemical signal?

Q7. Does poor water quality affect reproduction?